# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
# CHANGA

**B Tech (Electronics and Communication) Semester-III Examination Nov-2010(Regular)**
**EC – 204: Electronics Instruments & Measurement**

**Date: 30/11/2010, Tuesday** **Time: 10:30 a.m to 1:30 p.m** **Maximum Marks:70**

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**Instructions:**
1. **Attempt all questions.**
2. **Answer to the two sections must be written in separate answer books.**
3. **Figures to the right indicate *full* marks.**

## SECTION -I

| | | |
|---|---|---|
| **Q-1** | **Answer the following** | **10** |
| (a) | What is difference between accuracy and precision? | |
| (b | Explain in detail, the use of delay line in CRO. | |
| (c) | What is error? Explain different types of static error. | |
| (d) | What are balance conditions for AC bridge? | |
| (e) | Define Transducer; Give the difference between active and passive transducer. | |
| Q-2(a) | Draw the block diagram of CRO and explain function of each block. | 4 |
| (b) | Write a note on Dual Trace CRO. | 4 |
| (c) | For the following given data, calculate (i) Arithmetic Mean (ii) Deviation of each value (iii) standard deviation<br>Given $x_1 = 49.7$, $x_2 = 50.1$, $x_3 = 50.2$ $x_4 = 49.6$, $x_5 = 49.7$ | 4 |

**OR**

| | | |
|---|---|---|
| Q-2(a) | Explain construction and operation of Digital storage oscilloscope in detail. | 6 |
| (b) | Explain Function Generator with block diagram | 6 |

Q-3(a) What is limitation of Wheatstone bridge? Derive balance equation for Kelvin double bridge **5**

(b) The AC Bridge is in balance condition with following conditions: Arm AB, R= $450\Omega$ , Arm BC =$300\Omega$ in series with C=$0.265\mu F$, Arm CD is Unknown, Arm DA, R= $200\Omega$ in series with L=15.9mH. The oscillator frequency is 1Khz.Find the constants of arm CD. **5**

(c) Give the comparison between LED and LCD **3**

**OR**

Q-3(a) Explain wien bridge with figure and derive equation for frequency measurement **5**

(b) What is limitation of Maxwell bridge? Maxwell bridge is used to measure inductive impedance. The bridge constants at balance are : $C_1$= $0.01\mu F$, $R_1$= $470k\Omega$, $R_2$ = $5.1k\Omega$, $R_3$ = $1000k\Omega$. Find the series equivalent of unknown impedance. **5**

(c) What is difference between AC bridge and DC bridge? **3**

## SECTION-II

Q-4(a) Explain construction and operation of LVDT with neat figure **8**

(b) What do you mean by piezo- resistive effect? Derive equation of gauge factor for a bonded resistance wire strain gauge **6**

**OR**

(b) Explain Thermistor and Thermocouple in brief **6**

Q-5(a) Describe the working of a sweep frequency generator. What are the sweep errors? **5**

(b) Explain digital frequency meter with circuit **4**

**OR**

Q-5(a) Describe the Heterodyne Wave analyzer with the help of block diagram. **5**

(b) Describe digital measurement of time with basic block diagram. **4**

Q-6 **Write a short note on (Any Three)** 12

(a) BCD-to -7 Segment converters.

(b) Classification of Transducers.

(c) Sweep frequency generator

(d) Hay Bridge

(e) Frequency selective Wave analyzer